PHYS 2320 Lab Report

Title: Magnetic Force on a Current-Carrying Wire

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Date Performed: ___06/10/2026_______

Date Submitted: __06/16/2026________

Introduction

Purpose

The purpose of this experiment was to investigate the magnetic force exerted on a current-carrying conductor placed within a magnetic field. The experiment examined the relationship between current and magnetic force and verified that the magnetic force increases proportionally with current.

Theory

When a wire carrying an electric current is placed in a magnetic field, the wire experiences a force. The magnitude of this force is given by:

$F = ILB \sin(\theta)$

where:

F = magnetic force (N)

I = current through the wire (A)

L = length of wire in the magnetic field (m)

B = magnetic field strength (T)

θ = angle between the wire and magnetic field

For this experiment, the wire was positioned perpendicular to the magnetic field, so:

$\sin(90°) = 1$

Therefore:

$F = ILB$

This equation predicts a linear relationship between force and current when the magnetic field and wire length remain constant.

Experimental Setup

Equipment

• Current balance apparatus

• Digital balance

• Adjustable DC power supply

• Permanent magnets

• Connecting wires

• Current-carrying conductor loop

Procedure Summary

The magnetic force apparatus was assembled with the conductor loop suspended between the poles of a permanent magnet. The balance was zeroed before current was applied. Current through the conductor was increased incrementally from 0 A to 5 A. For each current value, the change in apparent mass was recorded. The measured mass change was then converted to force using:

$F = mg$

where $g = 9.8\ m/s^2$.

Lab Setup Photograph

Figure 1. Current balance apparatus used to measure magnetic force on a current-carrying conductor.

(Use the equipment photograph from the lab manual or your own setup photo if available.)

Data

Raw Data

Current (A)	Mass Change (g)
0.0	0.00
0.5	-0.31
1.0	-0.61

Current (A)	Mass Change (g)
1.5	-0.92
2.0	-1.24
2.5	-1.53
3.0	-1.82
3.5	-2.15
4.0	-2.46
4.5	-2.75
5.0	-3.06

Converted Force Data

Force was calculated using:

F = mg

Example:

m = 0.61 g

= 0.00061 kg

F = (0.00061)(9.8)

F = 0.00598 N

≈ 0.0060 N

Analysis

Sample Calculation

For 5.0 A:

Mass change = 3.06 g

m = 0.00306 kg

F = mg

F = (0.00306)(9.8)

F = 0.0300 N

Graph

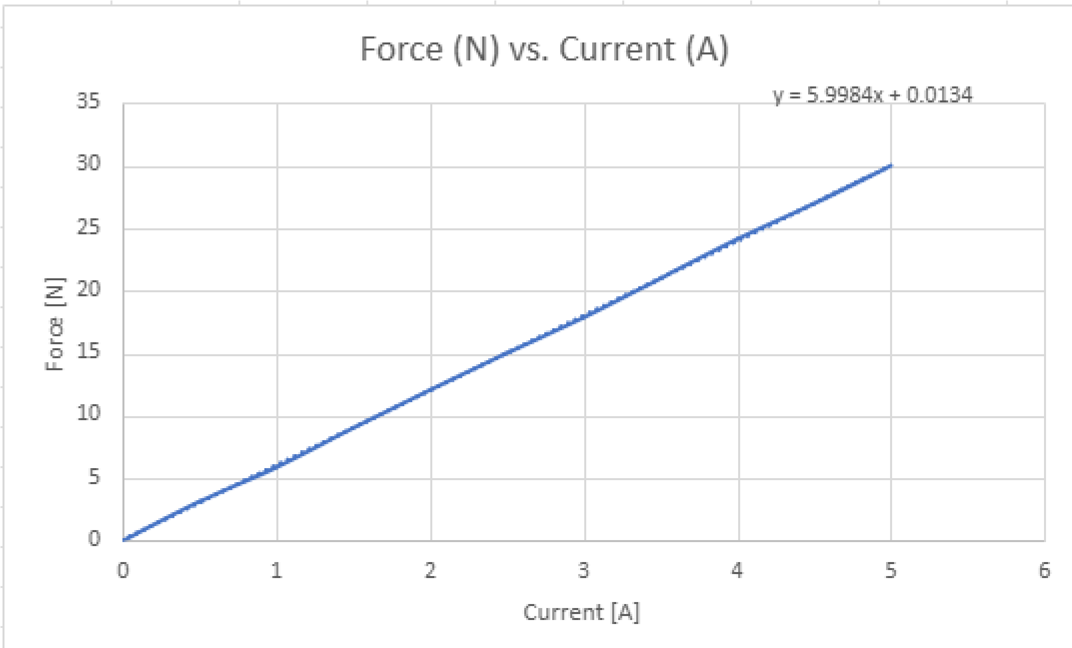


Figure 2. Magnetic force versus current.

The graph should display current on the horizontal axis and force on the vertical axis.

Interpretation of Results

The data show that magnetic force increased as current increased. The relationship appears approximately linear, which agrees with the theoretical equation:

$F = ILB$

Because the wire length and magnetic field remained constant throughout the experiment, force should be directly proportional to current. The graph demonstrates this expected trend. Small deviations from perfect linearity are likely due to measurement uncertainty and balance resolution.

Error Analysis

Several factors may have contributed to experimental uncertainty:

1. Limited precision of the digital balance.
2. Slight movement or vibration of the apparatus.
3. Variations in wire positioning within the magnetic field.
4. Electrical resistance causing small current fluctuations.
5. Reading and recording uncertainties.

Effect on Results

These factors could cause small deviations from the ideal linear relationship between force and current.

Suggested Improvements

• Use a balance with higher precision.

• Secure the apparatus more rigidly.

• Repeat measurements and average the results.

• Use a regulated power supply to maintain constant current.

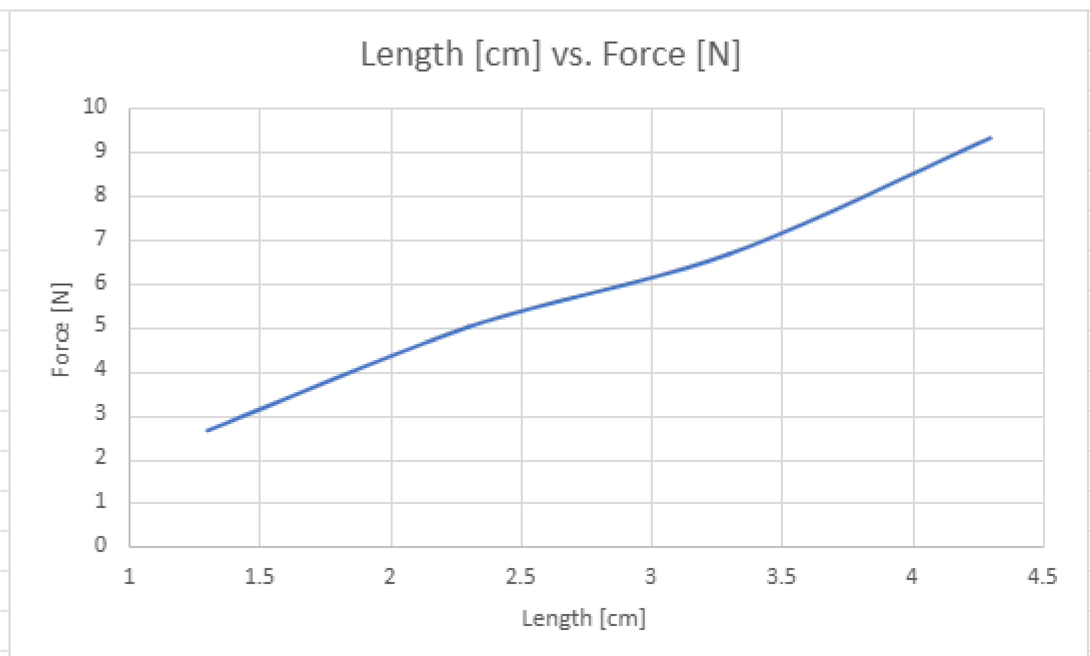
Conclusion

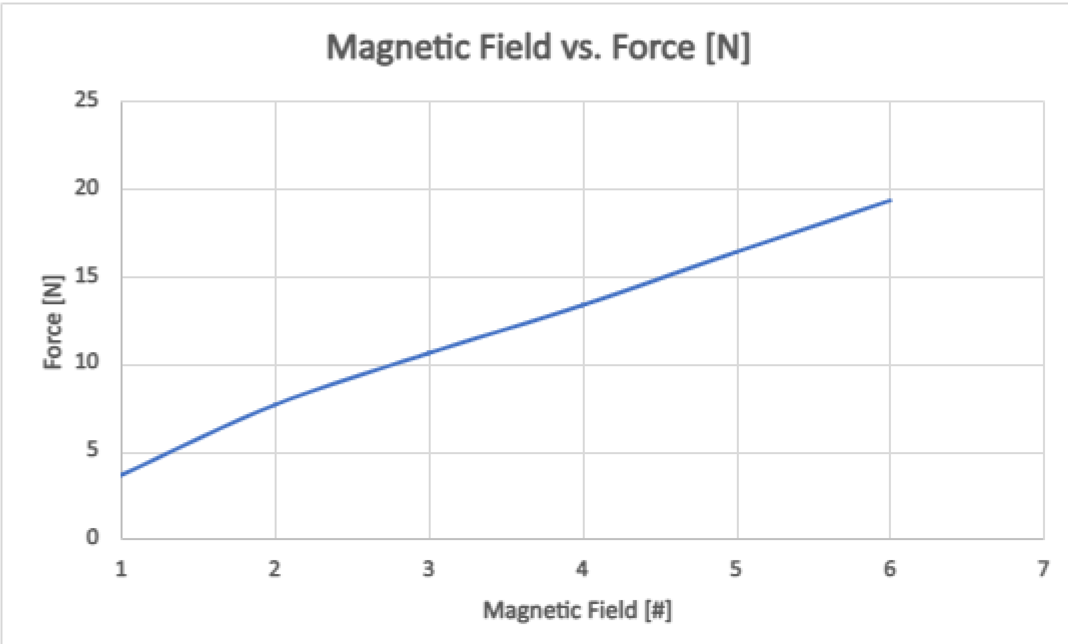
The purpose of this experiment was to study the magnetic force acting on a current-carrying conductor placed within a magnetic field. The experimental data demonstrated that magnetic force increased as current increased. The force-current graph showed an approximately linear relationship, consistent with the theoretical equation:

$F = ILB$

The results support the conclusion that magnetic force is directly proportional to current when the magnetic field strength and conductor length remain constant. Overall, the experiment successfully verified the relationship predicted by electromagnetic theory.

Appendix




Magnetic Field vs. Force [N]
Force [N]
Magnetic Field [#]
25
20
15
10
5
0
1
2
3
4
5
6
7